

01 INTRODUCTION

University campuses face significant challenges in managing classroom resources and maintaining academic integrity. Current manual methods lead to:

- **Reservation Confusion:** Double-booked rooms and administrative overhead.
- **Integrity Lapses:** Difficulty in scaling human supervision to detect subtle dishonest behaviors during modern exams.
- **Physical-Digital Disconnect:** Lack of real-time visibility into whether a room is physically occupied or free for study.

02 BACKGROUND

Background & Related Work

- **Smart Campus Initiatives:** Modernizing higher education through IoT and automated data collection to optimize resource use.
- **Related Work:** Existing systems often focus solely on specialized areas like biometric attendance or standalone cheating detection models without managing classroom utility.
- **The Improvement:** This project integrates visual validation with administrative scheduling to bridge the gap between digital data and the actual physical state of the classroom.

03 OBJECTIVES

Objectives

- **Automated Reservation:** Build a web interface for faculty to book classrooms and eliminate conflicts.
- **Real-time Checking:** Monitors booked spaces to provide students with live updates on classroom availability.
- **AI Exam Monitoring:** Develop a camera-based system to detect suspicious behavior (e.g., Phone usage, peeking) during assessments.
- **Privacy-First Design:** Process all video locally at the edge (Raspberry Pi/Local Host) to ensure no raw feeds are stored or broadcast.

04 METHODOLOGY

Methodology

- **Methodology:** Adopted the Agile System Development Life Cycle (SDLC) for iterative development and continuous stakeholder feedback.
- **System Architecture:** A Four-tier architecture (Presentation, Logic, Service, and Data).
- **Tools:**
 - **Frontend:** HTML, CSS, JavaScript, Jinja2.
 - **Backend:** Python with the Flask web framework.
 - **AI/Vision:** OpenCV and YOLO (You Only Look Once) for object detection and Gemini API.
- **Database:** SQL (SQLite) for storing reservations and logs.

04 RESULTS

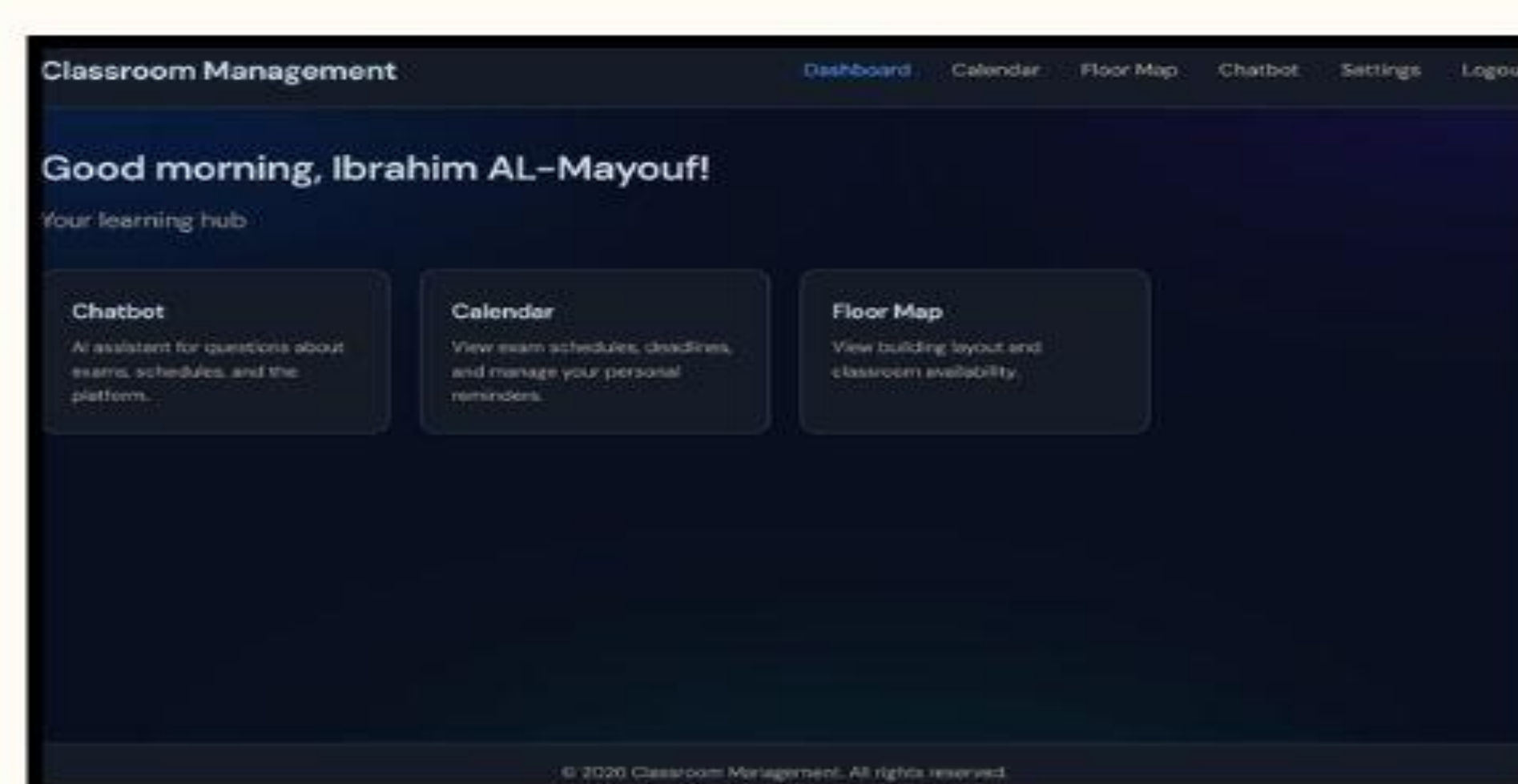
- **Performance:** Refactored modular architecture achieved extremely low latency (single-digit milliseconds) for system endpoints.
- **Model Accuracy:** Developed a working model capable of flagging peeking behavior with 74.3% mAP and phone detection with 69% mAP.
- **Functionality:** Successful integration of a new AI-powered Chatbot for natural language querying of schedules and room availability.
- **Testing:** Main user flows (Authentication, Admin Management, Calendar operations) behaved as intended during acceptance testing.

05 CONCLUSIONS & FUTURE WORKS

- **Conclusions:** The system successfully unifies classroom reservation, occupancy visibility, and AI integrity monitoring into one cohesive platform, reducing administrative friction and enhancing academic fairness.
- **Future Works:**
 - **Bilingual Support:** Implementing full Arabic/English localization.
 - **Dedicated Edge Hardware:** Transitioning to permanent Raspberry Pi installations for 24/7 monitoring.
 - **Institutional Scaling:** Optimizing the database to handle high-concurrency scheduling across the entire university.

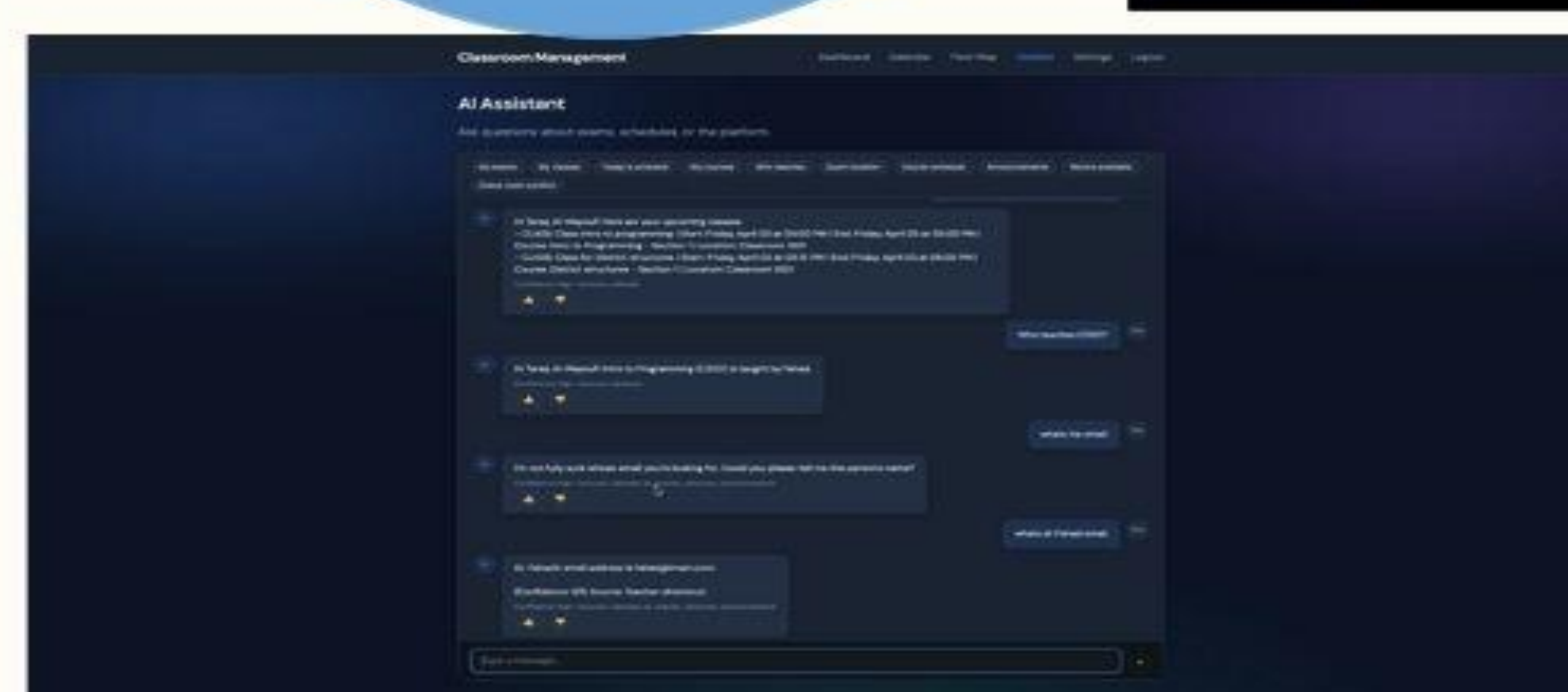
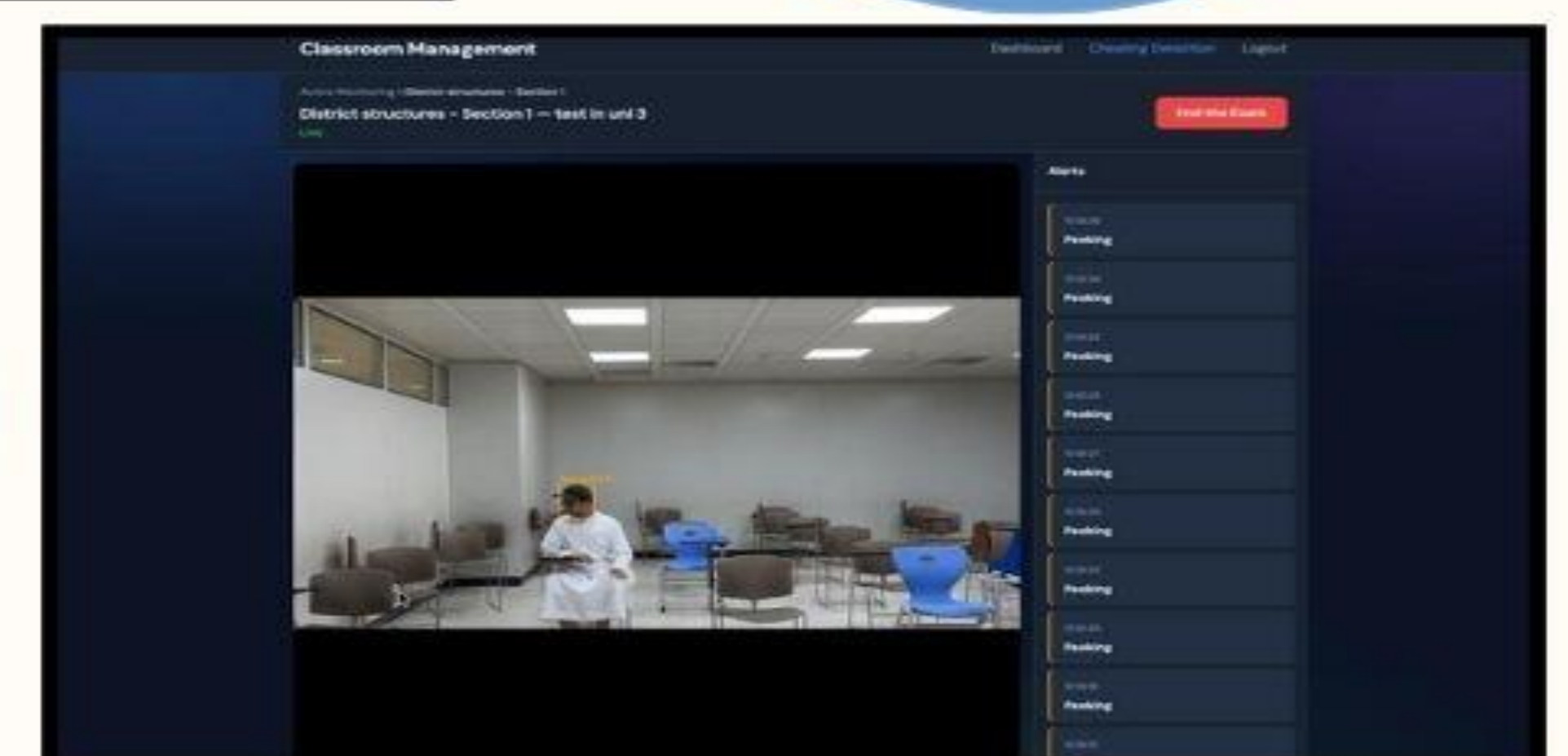
07 REFERENCES

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MAIN DASHBOARD

EXAM MONITORING



CHATBOT